Do Now · retrieval

Backtracking in computational solutions

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

When does a recursive call stop?

Q2

What is a search tree node?

Q3

Why may pruning reduce runtime?

Q4

Insert 12 into BST root10 with right child15 and left child5. Show the comparisons and position.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Make a decision — At each position choose the current item or skip it. Track index and running total.

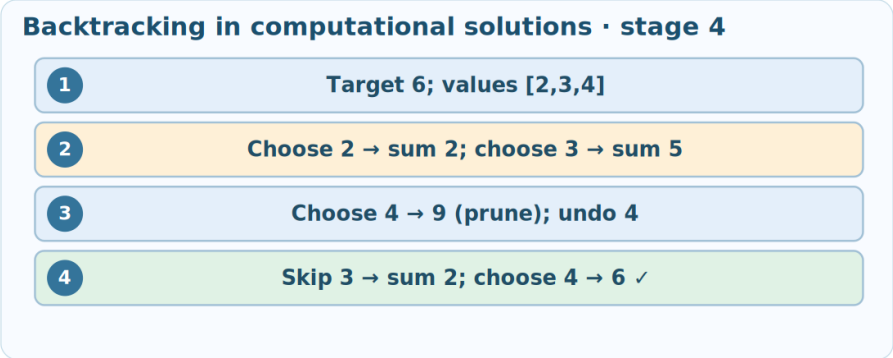
Core idea 2

Reject a dead end — When the sum exceeds the target (non-negative inputs) or the list ends, undo the choice.

Core idea 3

Backtrack — Try the other branch; success means a valid selection has been found. Exponential worst-case growth remains possible.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

backtracking — Explore choices and undo a failed path.

pruning — Discard an impossible branch.

search tree — Branches of possible decisions.

base case — Condition ending a recursive branch.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

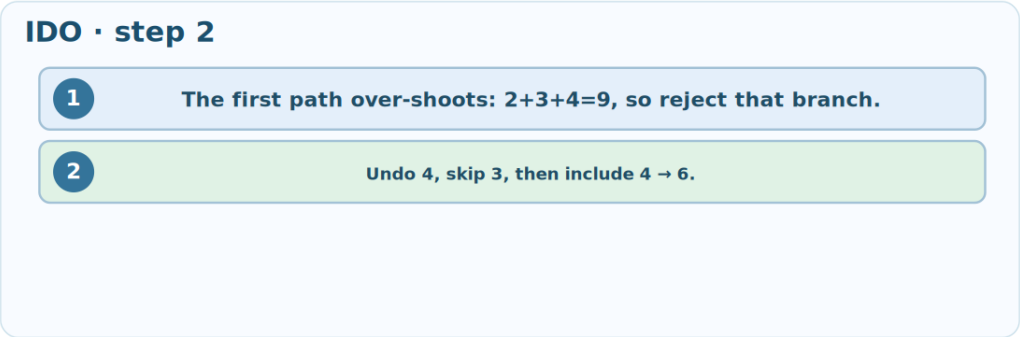
Problem

Find a subset totalling 6 from [2,3,4]. Trace the first failed path and successful branch.

Follow the worked method

1. Predict first — The first path over-shoots: 2+3+4=9, so reject that branch.
2. Explain the method — Undo 4, skip 3, then include 4 → 6.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first

Including 1 and 3 produces 4; adding 4 overshoots 5.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Explain choice, dead end and backtrack using [2,3,4] target6. [6]

Question 2

Write a recursive include/exclude Boolean function for subset sum with non-negative integers, including base cases. [8]

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Explain why backtracking is useful for 4 queens but not guaranteed polynomial time. Give one safe-placement rule. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Backtracking can require trying an earlier alternative after a dead end.

Use key vocabulary from memory

1. Explain backtracking in your own words.

2. Explain pruning in your own words.

3. Explain search tree in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.